

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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Platform	ServiceNow Flow Designer
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1. Introduction

1.1 Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1.2 Team Members

Name	Role	Responsibility
Rithish Nathan A	Team Leader	Overall coordination, solution design and flow logic review
Suriya Elango	Team Member	Requirement analysis and documentation
Vishal D S	Team Member	Catalog item, request form and portal front end
Thanigaivel E	Team Member	Testing, user acceptance testing and defect tracking
Prabhash Shukla R	Team Member	Flows, approval workflow and notifications

2. Project Overview

2.1 Purpose

The project automates the ordering of standard laptops on the ServiceNow platform using Flow Designer. It replaces a manual process of emails and free-text requests with a guided request, automatic validation and an approved-configuration recommendation. A procurement request is generated automatically, and anything outside the standard is detected and sent through an exception approval workflow.

The goals are to reduce manual procurement work, make laptop specifications consistent with the employee's job role, and shorten the time from request to order.

2.2 Features

- **Guided laptop request:** catalog form for employee, job role, CPU, RAM, storage, GPU, required software and budget.
- **Requirement validation:** mandatory fields and value ranges, including budget, are checked before processing.
- **Standard configuration matching:** the request is compared with the approved standard configurations.
- **Automated recommendation:** the matching approved configuration is recommended to the requester.
- **Procurement request generation:** created automatically for standard requests.
- **Non-standard detection and approval:** deviations are flagged and routed to an approver, who approves or rejects.
- **Notifications and audit trail:** requester and approver are notified, and every outcome is recorded.
- **Request report:** basic view of requests by job role, type, status and processing time.

3. Architecture

The solution is a low-code application on ServiceNow. The front end is a Service Catalog item shown in the Service Portal, the back end is a set of Flow Designer flows and subflows with supporting server logic, and the database is a set of ServiceNow tables.

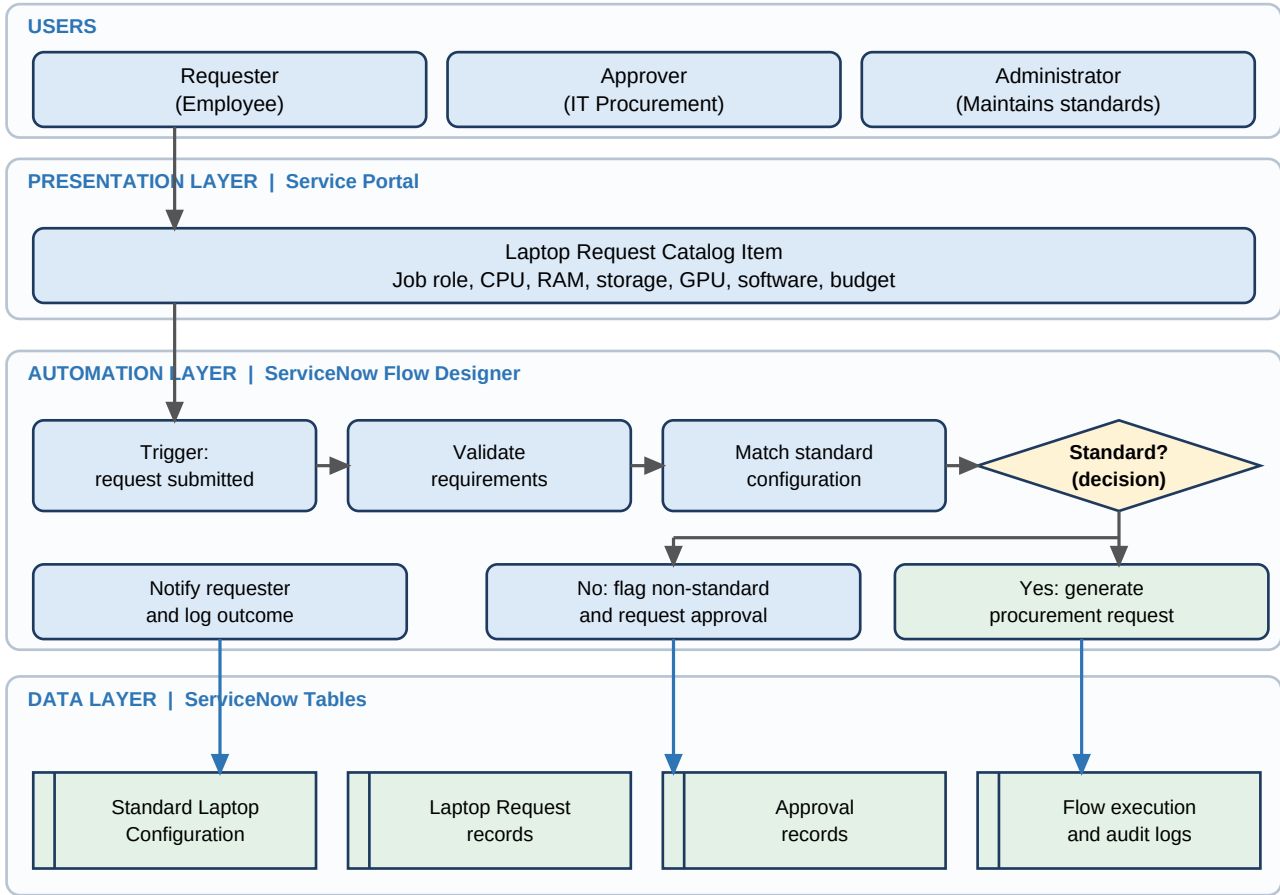


Figure 1: Solution architecture

3.1 Frontend

The front end is the **Laptop Request** catalog item displayed through the Service Portal. It collects the information the back end needs and gives immediate feedback.

Variable	Type	Purpose
Requested for	Reference (user)	Employee who will receive the laptop
Job role	Select box / reference	Drives the standard configuration lookup
CPU, RAM, storage, GPU	Select boxes	Requested specification
Required software	List collector	Software needed for the role
Budget	Currency	Checked against the allowed budget
Business justification	Multi-line text	Mandatory only for non-standard requests

Client-side logic (catalog client scripts and UI policies) marks mandatory fields, restricts invalid values and shows or hides the justification field, so most input errors are caught before the flow starts.

3.2 Backend

The back end is implemented in Flow Designer. The main flow is triggered when a laptop request is submitted and follows the logic in Figure 2.

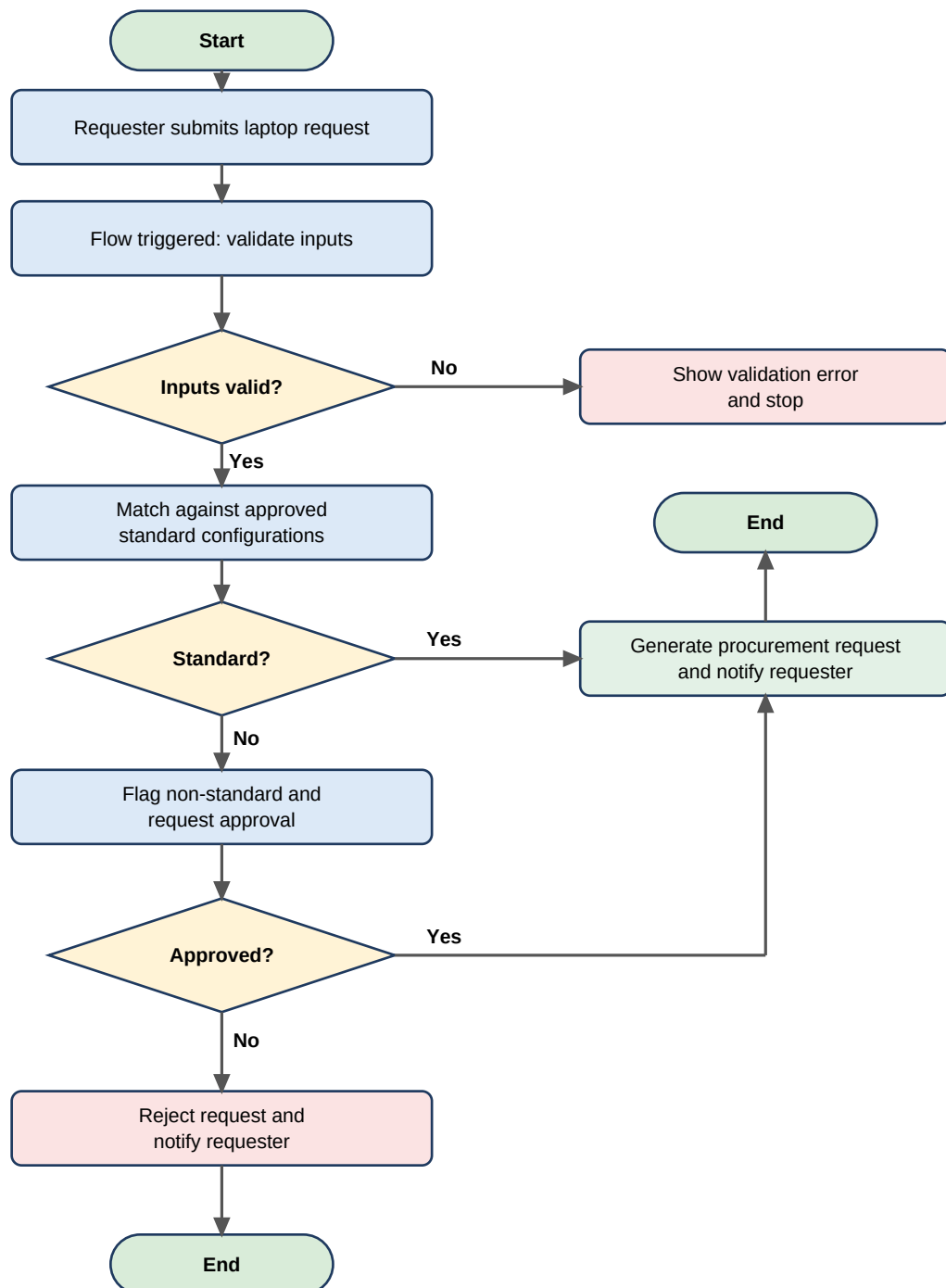


Figure 2: Laptop request automation flow

Component	Responsibility
Main flow: Standard Laptop Request	Trigger on submitted request; orchestrates validation, matching, decision, approval and generation
Subflow: Validate Requirements	Checks mandatory fields, value ranges and budget

Component	Responsibility
Subflow: Match Standard Configuration	Looks up the approved configuration for the job role and compares each specification
Subflow: Exception Approval	Creates the approval for non-standard requests and returns the decision
Action: Generate Procurement Request	Creates the procurement request with the approved details
Notifications	Messages to requester and approver at key stages

3.3 Database

Two custom tables hold the solution data. Table and field names are proposed and follow ServiceNow conventions.

Table	Key fields	Description
u_standard_laptop_config	config id, job role, CPU, RAM (GB), storage (GB), GPU, required software, maximum budget, active	Approved standard configuration for each job role; maintained by administrators
u_laptop_request	requested for, job role, CPU, RAM, storage, GPU, software, budget, matched configuration, request type (standard / non-standard), state, procurement request number	One record per laptop request, with the outcome of validation and matching
sysapproval_approver (platform)	approver, approval for, state, comments	Approval decision for non-standard requests

Each request is linked to the standard configuration it was matched against. Standard configurations are data, so a new role or specification is added by creating a record, without changing the flow.

4. Setup Instructions

4.1 Prerequisites

- A ServiceNow instance (for example a Personal Developer Instance) with an administrator account.
- Flow Designer, Service Catalog and Service Portal available (activated) on the instance.
- A modern web browser (Chrome, Edge or Firefox).
- Test user accounts for a requester and an approver.

4.2 Installation

- Sign in to the instance as an administrator and open the application scope used for the project.
- Create the roles, for example a procurement approver role, and assign them to the test users.
- Create the tables **u_standard_laptop_config** and **u_laptop_request** with the fields listed in section 3.3.
- Load the approved standard laptop configurations for each job role into the configuration table.
- Create the **Laptop Request** catalog item with its variables, client scripts and UI policies, and make it available in the portal.
- Build the subflows (Validate Requirements, Match Standard Configuration, Exception Approval) and the main flow in Flow Designer.

- Configure the approval action, the approver group and the email notifications.
- Set access control rules, then activate the flows and publish the catalog item.

Configuration values (system properties)

Property	Purpose
Approver group	Group that receives exception approvals
Procurement fulfilment group	Group that works the generated procurement requests
Budget currency	Currency used for budget validation

5. Folder Structure

ServiceNow stores a solution as platform records instead of files and folders. The application is organised as follows, with the client side and the server side described separately.

5.1 Client

Artifact	Description
Catalog item: Laptop Request	Request form shown in the Service Portal
Variables and variable set	Fields for job role, specification, software, budget and justification
Catalog client scripts	Field-level checks and show/hide behaviour
UI policies	Mandatory and read-only rules on the form
Portal page and widgets	Page that hosts the catalog item and the request status view

5.2 Server

Artifact	Description
Tables	u_standard_laptop_config and u_laptop_request
Flows and subflows	Main request flow, Validate Requirements, Match Standard Configuration, Exception Approval
Actions	Generate Procurement Request and supporting lookups
Notifications	Requester and approver email notifications
Roles and ACLs	Access rules for requesters, approvers and administrators
Reports	Request report by job role, type, status and processing time

6. Running the Application

The application runs on the ServiceNow instance, so no local server has to be started. To run it:

- Sign in to the instance as a requester and open the Service Portal.
- Open the **Laptop Request** catalog item, fill in the form and submit it.
- Open **Flow Designer > Executions** as an administrator to watch the flow run and to read the execution details.
- For a non-standard request, sign in as the approver, open the approval and approve or reject it.
- Check the generated procurement request and the notifications received.

7. API Documentation

The solution is used mainly through the portal, but its data can also be reached through the ServiceNow REST APIs. The endpoints below are those the solution relies on; table names follow section 3.3.

Method	Endpoint	Description
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Submit a laptop request through the catalog item
GET	/api/now/table/u_standard_laptop_config	Read approved standard configurations (filter by job role)
GET	/api/now/table/u_laptop_request/{sys_id}	Read a laptop request and its outcome
GET	/api/now/table/u_laptop_request	List requests (supports query parameters)
GET	/api/now/table/sysapproval_approver	Read approval records for a request

Parameters

Parameter	Used with	Description
sysparm_query	Table API GET	Filter, for example u_job_role=Developer^u_active=true
sysparm_fields	Table API GET	Comma-separated list of fields to return
sysparm_limit	Table API GET	Maximum number of records returned
sysparm_quantity and variables	Order now	Quantity and the catalog variable values

Example: submit a request

```
POST /api/sn_sc/servicecatalog/items/{sys_id}/order_now
{
  "sysparm_quantity": "1",
  "variables": {
    "job_role": "Developer",
    "cpu": "<cpu value>", "ram": "<ram value>",
    "storage": "<storage value>", "gpu": "<gpu value>",
    "budget": "<amount>"
  }
}
```

Example response

```
{
  "result": {
    "sys_id": "<request sys_id>",
    "number": "REQ00000000",
    "request_number": "REQ00000000",
    "request_id": "<request sys_id>"
  }
}
```

Example: read the standard configuration for a role

```
GET /api/now/table/u_standard_laptop_config
?sysparm_query=u_job_role=Developer^u_active=true&sysparm_limit=1
{
  "result": [ {
    "u_config_id": "<config id>", "u_job_role": "Developer",
    "u_cpu": "<cpu>", "u_ram": "<ram>", "u_storage": "<storage>",
    "u_gpu": "<gpu>", "u_max_budget": "<amount>"
  } ]
}
```

Values in angle brackets depend on the standard configurations loaded into the instance. Standard error responses (401 unauthorised, 403 forbidden, 404 not found) are returned by the platform.

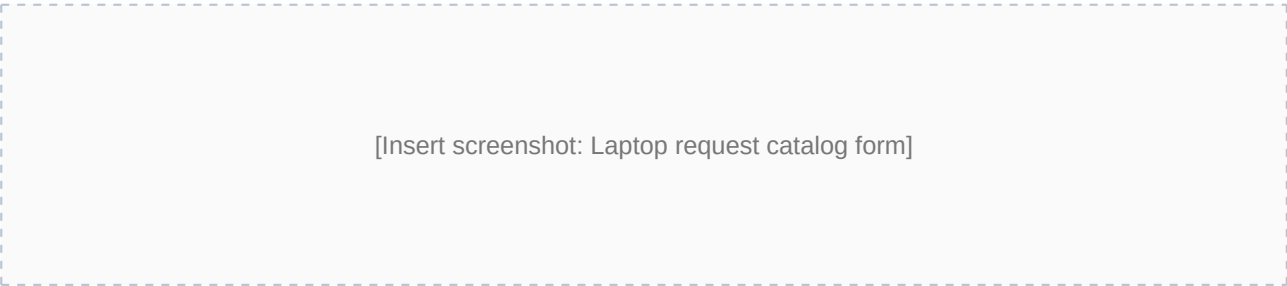
8. Authentication

Authentication and authorization are handled by the ServiceNow platform, so the solution does not implement its own login.

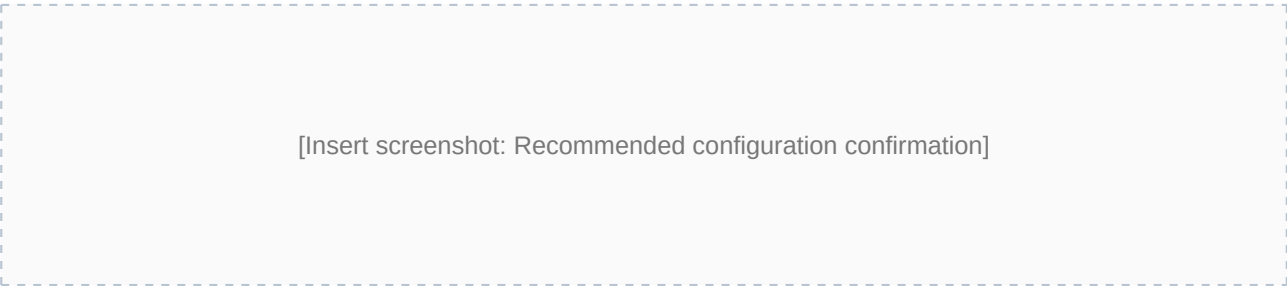
Aspect	Approach
Authentication (portal)	Users sign in to the instance with their organisational account or single sign-on; a session is created for the browser.
Authentication (API)	REST calls use basic authentication or an OAuth 2.0 bearer token issued by the instance.
Authorization	Role-based access control. Requesters can create and view their own requests; approvers can decide exception approvals; administrators maintain the standard configurations and flows.
Access control rules	ACLs on the tables restrict who can read, create, update or delete records and fields.
Auditability	Requests, approvals and flow executions are recorded with user and time.

9. User Interface

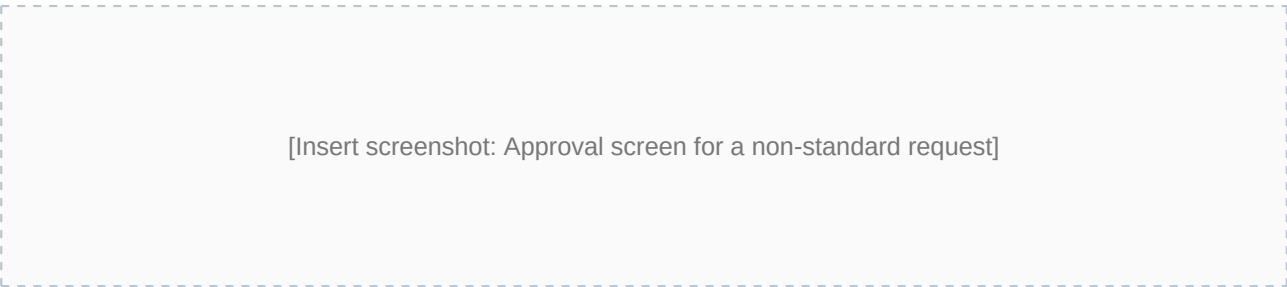
The user interface consists of the following screens. Screenshots are inserted in the frames.



Laptop request catalog form



Recommended configuration confirmation



Approval screen for a non-standard request

[Insert screenshot: Request status and report]

Request status and report

10. Testing

Testing combined functional tests, performance tests and user acceptance testing, using the flow execution log to confirm each result.

Test type	What was checked	Outcome
Functional	Input and budget validation, recommendation, non-standard detection, procurement request generation, flow execution	All cases passed
Performance	Response time (target under 3 seconds), concurrent requests, bulk requests	Targets met (about 2 seconds)
User acceptance	End-to-end scenarios by the team, 22 to 30 September 2026	All acceptance cases passed; defects fixed and re-tested

Detailed test cases and results are in the Functional and Performance Testing document and the UAT report.

11. Screenshots or Demo

[Insert screenshot: End-to-end flow: request submitted to procurement request generated]

Demonstration of the working solution

Item	Details
Demo link	[add demo video or link]
Repository	[add GitHub link]

12. Known Issues

No open critical or high-severity defects remained at the time of release. The points below are limitations to be aware of.

Item	Note
Standard configuration data	Recommendations are only as current as the data in the configuration table; administrators must keep it up to date.
Approver availability	A non-standard request waits until the approver decides.
Scope	Only standard laptop orders are handled; other equipment is not covered.
Platform dependency	The solution needs a ServiceNow instance with Flow Designer and Service Catalog.

13. Future Enhancements

- Vendor catalog integration for live pricing and availability.
- A virtual agent for raising laptop requests by chat.
- Analytics on orders and approvals to suggest updates to the standards.
- Extension to other IT equipment such as monitors, docks and mobile devices.
- Dashboards and budget forecasting for procurement managers.
- Asset management integration to track delivery and registration.